Alejandro Gomez De Mendieta

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Education

The University of Texas at Dallas — Richardson, TX
Bachelor of Science in Computer Science, Minor in Statistics — *Aug. 2025 – Dec. 2027*

Austin Community College — Austin, TX
Associate of Science in Computer Science; Associate of Science in Mathematics — *Aug. 2023 – Aug. 2025*

Experience

Software Engineer (Part-time) — Jan. 2022 – Jul. 2025
Ohel Technologies — *Austin, TX*

- Automated internal data and IT workflows for a five-person team and documented them in five SOPs, reducing recurring manual work by approximately five hours per week.
- Engineered biweekly ingestion pipelines integrating five external APIs with Elasticsearch, loading 100,000+ records in three months to support an AI application.
- Designed quota-aware ingestion logic around third-party rate limits and monthly usage caps to maintain reliable data collection.

Computer Science II Course Grader — Jan. 2026 – May 2026
University of Texas at Dallas — *Richardson, TX*

- Graded approximately 30 Java submissions per week for 71 students across two sections, maintaining a typical three-day turnaround against a seven-day policy.
- Provided rubric-aligned feedback on correctness, edge cases, code quality, and specification compliance to help students diagnose logic errors and improve implementations.
- Maintained shared notes on recurring mistakes and ambiguous edge cases to improve grading consistency across two graders.

Projects

Modern C++ HTTP Server | *C++23, Linux, POSIX sockets, CMake* — May 2026 – Present

- Built a from-scratch HTTP/1.1 server in C++23 using Linux/POSIX sockets, move-only RAII file descriptor ownership, layered TCP/HTTP abstractions, and method/path-based request routing.
- Implemented an incremental HTTP parser supporting fragmented socket reads, request-line/header/body validation, malformed-request handling, and response serialization; validated correctness with 111 Catch2 unit and process-level integration tests over real TCP sockets.
- Built a custom nanosecond-resolution benchmarking framework with per-iteration sampling and statistical analysis including mean, median, standard deviation, p95/p99 latency, throughput, and measurement-overhead correction across workloads of up to 100,000 iterations.
- Profiled parser workloads with Linux perf, traced ~85% of sampled self CPU to repeated buffer movement in `memmove`, and redesigned parsing around a cursor-based buffer, reducing byte-by-byte request parsing from ~414 ns to ~162 ns.

Homefull, HackUTD | *Python, NVIDIA Nemotron, DigitalOcean* — Nov. 8–9, 2025

- Deployed NVIDIA Nemotron on a DigitalOcean GPU droplet and built a Python backend that transformed crawled webpage content into structured JSON for local resource and event discovery.
- Implemented URL fetching, HTML parsing, event extraction, geocoding, and deduplication across the top five search results and pages crawled within each domain.
- Integrated an end-to-end discovery pipeline that returned event time and location information to the mobile application.

Technical Skills

Languages: C++, C, Python, Java, R
Systems and Tools: Linux/UNIX, POSIX sockets, TCP/IP, CMake, Git, CTest, GitHub Actions, Linux perf, Docker, Redis, Elasticsearch
Libraries: Catch2, nlohmann/json